

Plant phenolics as drug leads -- what is missing?

Low molecular weight phenols of plant origin are undoubtedly semiochemicals although not all of them can be easily classified as typical allelochemicals, which straightforwardly benefit the releaser. We have selected and surveyed this particular class of secondary metabolites, which shares high chemical reactivity with intrinsic biocompatibility and affinity for variety of molecular targets gained through evolution, because their suitability as prospective lead compounds for medicinal chemistry seems high but relatively unexplored. In particular, plant phenolics could be perceived as a natural product library, which contains privileged scaffolds, as evidenced by examples of endogenous phenols, phytochemicals containing aryl hydroxyl groups and phenolic synthetic drugs. It is postulated that application of bio-chemo-informatic tools to such library can be helpful in pulling out new drug candidates as well as in validating ADMET compatibility and suitability of the old ones. After short survey of structural diversity represented by plant phenolics, we focus on the compounds which either have obvious dietary significance or rich record of pharmacological studies, or both. It can be seen that apart from growing use of phytochemicals in dietary supplements, slow progress through clinical trials towards new drug registration is observed in that category of natural products. Such waste of resources on the way of transformation from renewable materials to high tech/high value products aimed for improved human healthcare is deplorable and should be reformed in name of sustainability. We attempt to answer the question why popular plant phenolics with well established health benefits and reasonably well recognized molecular pharmacology (such as: catechins, curcumin, resveratrol, quercetin and its glycosides, genistein, silymarin) have difficulties in attaining registered drug or even IND level.